James Aulds

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Professional Summary

Accomplished and adaptable Senior Data Engineer with proven expertise in **Microsoft Azure** ecosystems, including **Azure Data Factory** and **Azure Synapse Analytics**. Demonstrates strong capabilities in **data engineering** leadership, driving delivery excellence and collaborating effectively across cross-functional teams to architect scalable, reliable data pipelines and solutions that meet evolving business needs. Adept at integrating complex **SQL** solutions and implementing modern data architectures that empower informed decision-making, while fostering continuous learning and mentoring teams for sustained success.

Skills

Programming Languages

SQL · Python · Scala · Java · C# · T-SQL · DAX · R · PowerShell · JSON

Frameworks & Libraries

Apache Spark · .NET Framework · Entity Framework · Azure SDK · Power BI Embedded · ML.NET · TensorFlow · Pandas · Scikit-learn · PyTorch · Azure Databricks · MLflow · Azure Synapse Analytics Notebooks · .NET Core · Azure Functions

Tools & Platforms

Microsoft Azure · Azure Data Factory · Azure Synapse Analytics · Azure Databricks · Power Platform · Azure SQL Database · Microsoft Dynamics 365 · Azure DevOps · Git · Visual Studio · SQL Server Management Studio (SSMS) · Azure Active Directory · OAuth 2.0 · REST APIs · Microsoft Power BI

Databases

Azure SQL Database · SQL Server · Azure Data Lake Storage · Cosmos DB · PostgreSQL · MySQL · Oracle · MongoDB · Redis · Azure Synapse Analytics (Dedicated SQL pool) · Azure Blob Storage

DevOps & CI/CD

Azure DevOps Pipelines | GitHub Actions | Terraform | ARM Templates | Jenkins | Docker | Kubernetes | Ansible | Helm | SonarQube | PowerShell scripts

Other Skills

Data Orchestration | Dimensional Modeling | Data Warehousing | Data Integration | Monitoring & Alerting | Metadata Driven Pipelines | Data Governance | Performance Tuning | Cloud Security Architecture

Certifications

- Microsoft Certified: Azure Data Engineer Associate
- Microsoft Certified: Azure Solutions Architect Expert
- Microsoft Certified: Azure Fundamentals
- Microsoft Certified: Azure Security Engineer Associate

Work Experience

Senior Data Engineer - Johnson & Johnson

08/2022 - Present

Led backend development integrating **Microsoft Azure** services with **Azure Data Factory** and **Azure Synapse Analytics** to deliver AI-driven surgical video analysis features, improving operational efficiency using advanced **SQL** data orchestrations and real-time data processing pipelines.

Architected modular **data engineering** pipelines leveraging **Azure Data Factory** and **Azure SQL Database** to automate the ingestion and transformation of surgical video metadata, enabling rapid highlight reel generation with high data quality and reliability.

Collaborated closely with cross-functional teams to merge clinical data via **Azure Synapse Analytics**, enhancing preoperative planning and postoperative monitoring by integrating extensive patient health records and AI insights.

Developed secure, scalable backend services implementing OAuth 2.0 authentication protocols combined with **Azure Active Directory** to enable telepresence collaboration features across distributed surgeon teams worldwide.

Designed and optimized complex **SQL** dimensional models and star schemas within **Azure Synapse Analytics**, yielding accelerated queries for surgical workflow and analytics dashboards, boosting accessibility and decision-making for clinicians.

Implemented AI video annotation workflows alongside **Azure Databricks** to analyze surgical footage metadata, generating actionable insights embedded into the platform and integrated with real-time data orchestration pipelines in **Azure Data Factory**.

Applied best practices for cloud security and compliance governance within **Azure**, ensuring sensitive patient data encryption and secure data transfers aligned with healthcare industry standards.

Executed continuous integration and deployment automations using **Azure DevOps Pipelines**, enabling rapid iteration of backend features and maintaining high system uptime in critical surgical environments.

Mentored junior engineers in the adoption of **data engineering** best practices and advanced **SQL** optimization techniques within the **Azure Synapse Analytics** environment.

Enabled telemetry and logging solutions using **Azure Monitor** and custom alerting pipelines to maintain operational excellence and proactive performance monitoring of the AI-driven analytics platform.

Integrated third-party APIs with OAuth 2.0 authorization to pull external surgical data into **Azure Data Factory** pipelines, standardizing JSON data into relational formats in **Azure SQL Database**.

Actively participated in design sessions and documentation efforts to capture intricate data flow diagrams and maintain thorough architectural documentation for enterprise-level data solutions.

Spearheaded the transition from legacy data systems to **Azure**-based lakehouse designs combining **Azure Data Lake Storage** with **Azure Synapse Analytics** to optimize data retrieval times and storage cost.

Coordinated with business intelligence teams leveraging **Power BI** to build real-time reporting dashboards driven by the backend pipelines and dimensional models developed on **Azure SQL Database**.

Utilized modern data orchestration patterns in **Azure Data Factory** to ensure idempotent, metadata-driven workflows supporting automated surgical video data processing at scale.

Championed security architecture reviews and implemented role-based access controls using **Azure Active Directory**, safeguarding PHI within the Polyphonic™ digital ecosystem.

Engaged with project management and clinical stakeholders to refine data requirements and translate complex surgical procedures into actionable data engineering tasks within the **Azure** cloud environment.

Optimized infrastructure as code deployment with **Terraform** integrating **Azure** resource provisioning alongside CI/CD pipelines to streamline updates and secure configurations.

Enhanced real-time alerting and anomaly detection in equipment and surgical process workflows using advanced **SQL** analytics within **Azure Synapse Analytics**.

Collaborated on research papers and internal knowledge sharing focused on leveraging **Azure Databricks** and AI methodologies to advance healthcare data innovation.

AI Engineer - Walmart

01/2020 - 07/2022

Architected and implemented scalable **data engineering** solutions on **Microsoft Azure** leveraging **Azure Data Factory** and **Azure SQL Database** to ingest and process merchant data for the AI-powered assistant “Wally,” enabling automated data handling and performance diagnostics.

Developed complex **SQL** transformation logic and dimensional modeling on **Azure Synapse Analytics** that supported real-time analytics and operational querying for Walmart’s merchandising decisions.

Integrated multiple external API sources secured with OAuth 2.0 into **Azure Data Factory** pipelines, normalizing JSON data structures and optimizing ingestion efficiency to support advanced AI functionality.

Collaborated with cross-disciplinary teams including Azure cloud administrators, business analysts, and BI developers, delivering end-to-end Microsoft data solutions aligned with Walmart’s operational needs.

Designed modular, metadata-driven orchestration workflows in **Azure Data Factory** ensuring automated, reliable, and scalable data flows that evolved with business requirements and AI iteration cycles.

Led performance tuning efforts of **Azure SQL Database** relational structures and indexing strategies, substantially reducing query response times and enhancing real-time metric reporting.

Delivered hands-on mentoring for junior data engineers focused on **Microsoft Azure** tooling, **data engineering** best practices, and advanced **SQL** optimization techniques within the retail domain.

Employed **Azure DevOps** CI/CD pipelines integrated with **Terraform** to automate consistent cloud infrastructure deployments supporting the AI assistant platform’s reliability and uptime.

Constructed comprehensive documentation and training materials for client workshops and team knowledge sharing to promote standardized practices across the data engineering team.

Implemented data security and role-based access control leveraging **Azure Active Directory** to protect sensitive merchant data processed within the AI system.

Oversaw end-to-end lifecycle management of AI data pipelines from ingestion, transformation, quality validation, and storage in **Azure Blob Storage** and **Azure Data Lake Storage**.

Architected data integration strategies using **Azure Synapse Analytics** to combine transactional data with operational metrics, enabling multifaceted business intelligence solutions driven by machine learning insights.

Applied dimensional modeling techniques and star schema design in **Azure SQL Database** to facilitate straightforward report generation and consistency in analytics outputs.

Developed monitoring dashboards using **Power BI** connected directly to **Azure Synapse Analytics** queries, providing stakeholders with near real-time insights into merchandising KPIs.

Integrated cloud-native security features and audit logging into data orchestration processes to maintain compliance with corporate and industry data standards.

Leveraged **Azure Databricks** notebooks for iterative data exploration and machine learning model data preparation supporting the AI assistant’s development lifecycle.

Directed the automation of operational support queries via AI-driven data workflows, combining **Azure Data Factory** orchestration with backend **SQL** transformations for self-service capability enhancements.

Coordinated cross-team collaboration to embed business intelligence insights within automated troubleshooting processes, enabling faster decision-making and issue resolution.

Enhanced data pipeline robustness by integrating error handling, retry mechanisms, and alerting through **Azure Monitor** and custom logging frameworks.

Championed methodologies for scaling data engineering solutions that harmonize metadata-driven pipeline construction with **Azure** cloud best practices.

Machine Learning Engineer - Caterpillar Inc.

07/2017 - 12/2019

Delivered advanced **data engineering** and machine learning solutions on **Microsoft Azure** infrastructure leveraging **Azure Data Factory**, **Azure Synapse Analytics**, and **Azure Databricks** for the Cat® Predictive Maintenance Platform to reduce unplanned equipment downtime.

Engineered complex **SQL** queries and dimensional models on **Azure SQL Database** and integrated IoT sensor data through **Azure Data Factory** pipelines, enabling real-time monitoring of equipment health with predictive analytics capabilities.

Utilized **Azure Data Lake Storage** as a scalable repository for high-velocity sensor data streams and performed data normalization and flattening of JSON telemetry data for efficient analysis.

Led architectural design and deployment of machine learning workflows in **Azure Databricks**, incorporating Spark-based data processing to forecast equipment failure and generate automated maintenance alerts.

Collaborated with cloud administrators and cross-functional teams to optimize data orchestration and monitoring pipelines using **Azure Data Factory**, increasing reliability and scalability of the predictive platform.

Implemented secure OAuth 2.0 API connectors within data pipelines to ingest external industrial data along with internal telemetry, standardizing inputs for unified analytics across segments.

Developed reusable metadata-driven pipelines to automate data movement, transformation, and quality checks, ensuring consistent accuracy in predictive insights delivered to field technicians.

Spearheaded optimization of **SQL** performance on large datasets within **Azure Synapse Analytics**, reducing query latency and enhancing the responsiveness of monitoring dashboards.

Enforced cloud security architecture best practices, integrating role-based access control with **Azure Active Directory** and data encryption protocols to protect critical operational data.

Provided leadership and mentorship to junior engineers, fostering expertise in Microsoft data services and best practices in building extensible, scalable data platforms.

Collaborated in cross-disciplinary design sessions, documenting architectural decisions and maintaining comprehensive system diagrams to support scaling and future improvements.

Automated deployment workflows via **Azure DevOps** pipelines with infrastructure as code templates developed in **Terraform** for consistent environment provisioning.

Integrated business intelligence tools such as **Power BI** for visualizing real-time equipment health metrics and maintenance forecasts, facilitating proactive customer interventions.

Designed and developed dimensional star schema models to streamline reporting and leverage predictive analytics for operational decision support.

Enhanced data quality and system resilience through proactive logging and alerting mechanisms implemented with **Azure Monitor** and custom telemetry.

Led efforts to combine data warehousing and lake storage paradigms within the Azure ecosystem to optimize cost-efficiency and data accessibility.

Applied advanced Spark analytics in **Azure Databricks** notebooks to test and refine machine learning model features, improving predictive accuracy.

Participated actively in client technical reviews and knowledge transfer sessions to promote adoption of robust data engineering standards.

Delivered solutions utilizing metadata-driven pipeline techniques ensuring maintainability and ease of feature enhancement across evolving client requirements.

Championed continuous improvement initiatives integrating feedback and iterative enhancements within the cloud-native data engineering frameworks.

Education

Master's Degree in Computer Science

Cabrini University, 2021

Bachelor's Degree in Computer Science

Cabrini University, 2016